

Group Assessment Coversheet

To be attached to the front of the assessment.

Campus: Tygervalley Campus

Faculty: BSc IT Software Engineering

Module Code: ITDMA3-22

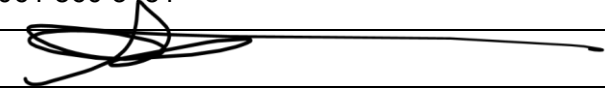
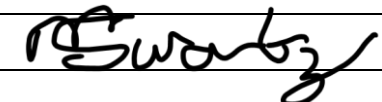
Group: S10

Lecturer's Name: Azeez Olasaji

Indicate	Yes	No
Plagiarism report attached	☒	☐

Declaration:

We declare that this assessment is our own original work except for source material explicitly acknowledged. We also declare that this assessment or any other of our original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. We are aware of the AI policy and acknowledge that we have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. We also declare that we are aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

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Question 1: Introduction

Enhancing Client Visualization in Interior Design through Android-Based Augmented Reality Tool

Augmented reality¹ (AR) has rapidly emerged as a transformative technology in information technology (IT) and design, enabling digital 3-dimensional (3D) content to be overlaid on the physical world to improve client understanding of spatial environments (Chen, et al., 2019). In interior design, AR applications allow virtual layouts and furniture to be placed in real spaces, aiding users to visualize spatial relationships and dimensions that would otherwise require imagination or costly mockups (Ahsani, et al., 2025). Major trends include smartphone-based AR (ARKit/ARCore) that leverage cameras and sensors to display interactive 3D models of décor and furniture in real time. This has been driven by industry solutions such as Kubity Go and Wayfair Decorify, as well as research prototypes. Key challenges persist: accurate visualization in complex environments and creating intuitive interfaces for non-technical clients (Iqbal, et al., 2022). Despite advances in AR technology, current tools for interior design primarily focus on designer workflows—such as IKEA Place, which enables users to virtually place furniture in their space (Kowalczyk, et al., 2021). This represents a gap in truly user-centric mobile solutions. Supporting this observation, (Chang, et al., 2019) demonstrated that mobile AR improved the confidence and satisfaction of users when arranging furniture. Interior designers still rely on traditional 2-dimensional (2D) plans and schematics, which limit client comprehension and feedback. This research is set against these trends and gaps: while AR can bridge the gap between plan and personal experience (Hilken, et al., 2022), there is limited research on Android-based AR apps tailored for clients, an important perspective given the prevalence of Android devices (Zamłyńska, et al., 2022). This gap is critical because the core challenge in interior design remains the communication barrier between designers and clients—where current visual methods often fail to convey spatial intent, leading to misinterpretation and costly revisions.

Problem statement:

One persistent challenge to the practice of interior design is the disconnect between the conceptual illustrations of designers and the capacity to accurately interpret proposed design solutions of the client. Clients tend to misunderstand two-dimensional drawings and renderings, which creates miscommunication and results in costly project changes (Vijaya Lakshmi, et al., 2025). Research studies have found that clients prefer augmented reality visualizations because they allow spatially accurate and realistic placement of design elements in real contexts, allowing for stronger comprehension, and better-informed decision-making.

Although AR has the potential of addressing the understanding of clients with interior designs, the application of these technologies on mobile devices, especially across the Android ecosystem, continues to be limited by practical usability and more importantly, performance. Practical constraints are due to variances in hardware, camera quality, and performance availability across devices that ultimately do not enable seamless or dependable AR experiences. Therefore, these technical constraints need to be resolved to ensure the practical development of AR solutions that could fundamentally improve client visualization and communication in tandem with interior design projects (Lohakare, et al., 2024) (Vijaya Lakshmi, et al., 2025).

Research questions:

- **Primary Research Question:**

How can an Android-based AR tool be optimized to improve client-designer communication in interior design projects?

¹ All further references to “augmented reality” in this research will be abbreviated as AR

- **Sub-Questions:**

1. What are the key usability and performance limitations of current mobile AR applications that hinder client engagement in interior design, particularly on low- to mid-level Android devices?
2. Which user-interface (UI) and user-experience (UX) design principles make AR tools more intuitive for clients unfamiliar with design software?
3. How do differences in Android device hardware affect the accuracy and smoothness of AR experiences?
4. What evidence from existing studies demonstrates that AR improves client understanding of interior design concepts compared to traditional two-dimensional (2D) plans?

Research objectives:

- **The General Objective:**

The general objective of this research proposal is to design, develop, and evaluate an Android-based AR application that enhances client visualization and engagement in interior design projects by addressing usability and performance challenges that are commonly associated with mobile AR platforms.

- **Specific Objectives:**

1. To analyse existing AR applications used in interior design and identify key usability and performance limitations on Android devices.
2. To design an Android-based AR application aimed at improving client understanding and interaction with interior design concepts.
3. To optimize the performance and usability of the developed AR application for a range of Android hardware specifications.
4. To provide practical recommendations for further enhancements to mobile AR tools that support client-centered interior design visualization.

Justification for the study:

In interior design practice, the visualization gap between designers and clients has long stunted project efficiency. Traditional 2D drawings, which are limited by their spatial representation often lead to client misunderstanding of design proposals, thus triggering frequent design changes and cost overruns (Mohammed, et al., 2021). AR technology presents an innovative solution that bridges this gap by overlaying 3D virtual elements onto real environments, significantly enhancing clients depth of understanding of spatial layouts and boosting decision making confidence.

However, implementing this technology on the Android platform faces significant challenges. As the mobile operating system with the highest global market share, as illustrated in Figure 1, Android devices suffer from hardware fragmentation. This means there are variations in processor performance and camera accuracy across different models (Sun, 2023). As a result, existing AR tools often experience interactive lag and positioning inaccuracies on mid to low end devices, which severely restricts the adoption of this technology (Lee, 2020). Existing literature acknowledges both the benefits on AR in interior design and the technical constraints posed by Android devices but

lacks targeted solutions for optimizing AR applications specifically for interior design context (Monika A. Ganpate, 2015).

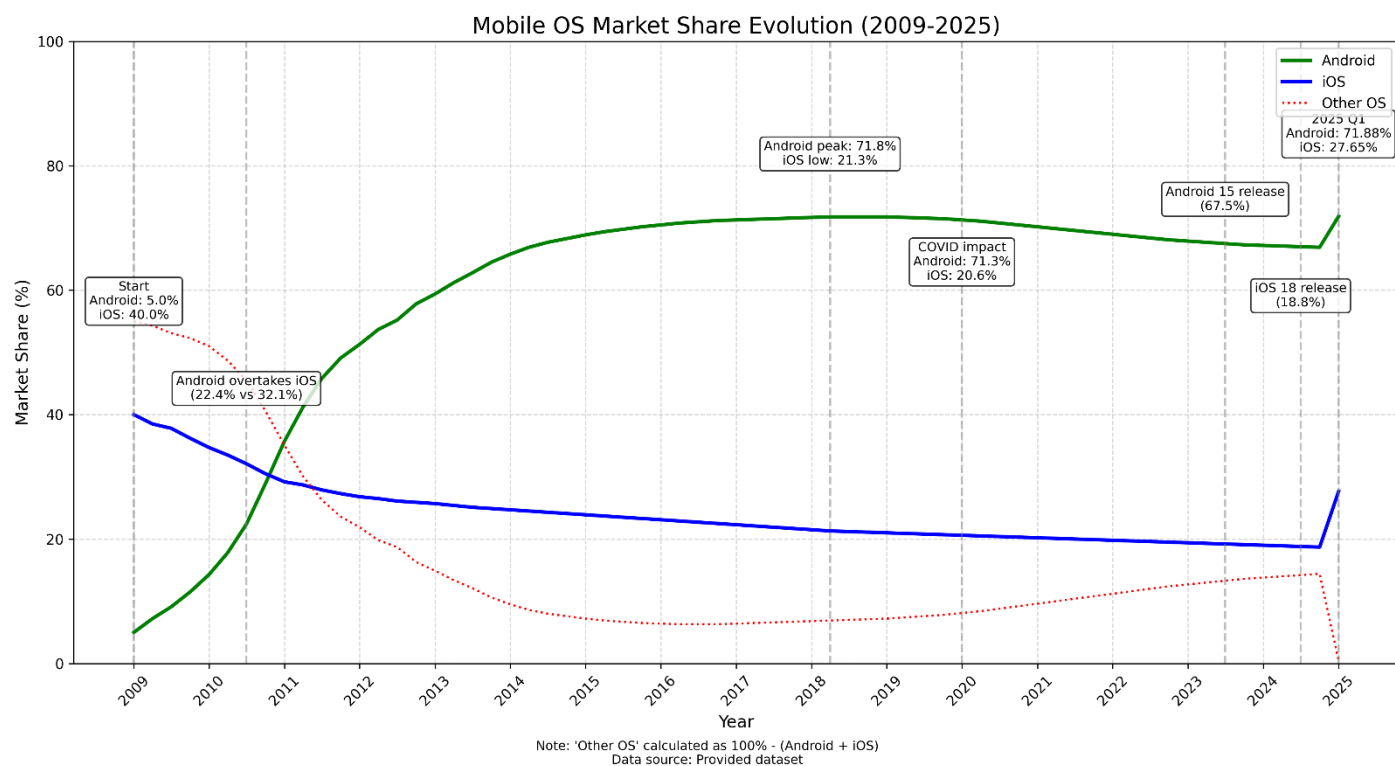


Figure 1: Mobile Operating System(OS) market share evolution (2009-2025). Reference: Statista

Currently research on Android based AR in interior design fail to address these core gaps:

- Diversity in Hardware: Most research focuses on iOS or high-end devices, ignoring performance optimization for mid-range Android models, despite Android's dominant market share (Sun, 2023).
- User-centered design shortage: Existing tools prioritize designer workflows over client-centric interactions.

This research fills these gaps by proposing a tailored AR solution that enhances client visualization, streamlines designer-client communication, and reduces project iterations. For interior design practitioners, this translates to improved client engagement and satisfaction. Technology and design policymakers can leverage these insights to establish hardware compatibility standards, guiding manufacturers to optimise configurations for AR applications thereby boosting industry efficiency(Wilcox, 2025).

Theoretically, the research:

- Develops a roadmap for developing successful AR applications on resource-constrained platforms (Santhosh, et al., 2023).
- Advances knowledge of mobile AR optimization by integrating user-centred design (UCD) principles with technical implementation (Santhosh, et al., 2023).
- Bridges the divide between the theoretical advantages of AR and its practical realization on Android devices, fostering more effective interior design collaboration.

By resolving Android specific AR challenges, the research aligns with broader IT objectives to deliver scalable, user friendly technologies. It demonstrates how targeted optimization of fragmented hardware ecosystems can drive innovation, ultimately improving project outcomes and client experiences. This research will position AR as a viable, accessible tool for industry wide adoption reinforcing the IT sectors role in enabling efficient client focused design solutions.

Question 2: Literature Review

Literature Understanding and Gaps

Augmented reality (AR) is shown to greatly enhance the ability of clients to visualize interior design concepts by overlaying virtual décor onto real spaces. Studies note that AR “bridges the gap between virtual and the real environment,” allowing users to place true-to-scale 3-dimensional (3D) models in their rooms and thus make better decisions (Revathy, et al., 2024; Liu, et al., 2025). For example, (Revathy, et al., 2024) describe an AR system where users can “virtually place furniture in their living spaces ... by accurately visualizing their desired interior design”. Similarly, (Liu, et al., 2025) emphasize that AR “provides an immersive interactive experience, enabling users to make more intuitive decisions during the design process”. AR tools support detailed client visualization: clients can observe and walk around virtual décor, adjust lighting or colors, and preview layouts in real time, helping clients “make informed purchase decisions and reducing common mistakes” (Revathy, et al., 2024).

AR enables clients to preview items at true scale, improving confidence (R, et al., 2025). For instance, an Android AR app allowed users to load realistic 3D furniture models into their own environment, resulting in “increases in [decision-making] satisfaction and a drop in return rates” (Hui, 2015). Retailers increasingly leverage AR to bridge online and in-store shopping experiences. Home Depot’s Paint Visualizer app, for example, allows users to overlay paint colours onto their own walls in real time, helping them preview results without a store visit (Yoo, 2020). IKEA Place similarly enables users to place true-to-scale virtual furniture in their spaces; research shows this enhances immersion, ease of use, and purchase intention (Nhan, et al., 2022). A study on mobile AR shopping (Yoo, 2023) confirmed that interactive 3D features, such as rotation and zoom, increase perceived usefulness and telepresence, which significantly boosts behavioral intention to purchase. Additionally, (Hoffman, et al., 2022) found that AR enhances immersion, resulting in greater enjoyment, perceived usefulness, and purchase intent, particularly for users with less prior media experience.

Overall, current literature consistently reports that AR enhances client visualization and engagement. By overlaying décor, AR helps clients “conceptualize their design, and make proper decisions” (Revathy, et al., 2024). Empirical findings confirm with examples such as SLAM-based mobile AR prototypes that successfully promoted interior design products by giving users a realistic preview (Ramadhan, et al., 2021).

Despite these advances, significant gaps remain, especially regarding mobile, Android-focused AR solutions and rigorous user-centered studies. A recent review by (Liu, et al., 2025) highlighted several technical challenges in AR interior design: 3D interaction complexity, environmental lighting variations, and cross-platform compatibility. For example, designing natural touch and movement controls in AR remains difficult, and fluctuating light in a room can distort how virtual objects appear. Moreover, many existing AR tools are platform specific. A chief concern is ensuring an app works on both iOS and Android devices; as (Liu, et al., 2025) noted, “AR solutions must operate across different devices and operating systems” to meet diverse user needs.

Beyond technical hurdles, literature points out conceptual gaps. Most AR research focuses on educational or desktop-based design, with fewer studies on client-centric mobile AR. For example, the Bahrain interior design study assessed AR adoption in offices, not end-user applications (Hui, 2015). While industry applications (such as IKEA Place or Wayfair’s AR tools) exist, peer-reviewed evaluations of such mobile AR retail applications are lacking. (Biswas, 2022) is among the few who explicitly built an Android AR prototype for furniture browsing. His work notes the broad compatibility of ARCore, but it remains a rare Android-centric implementation. In general, literature still lacks user studies on Android AR interior apps: for instance, (Mohammed, et al., 2021) conducted a usability

study of an AR interior app to optimize the interface, but such work is isolated. In conclusion, the weakest aspects of AR interior design research are the limited focus on mobile (especially Android) deployment and the need for more client-focused evaluations.

Successes and Prototypes

Researchers have begun to address these gaps with diverse AR prototypes and studies. One such prototype is mobile AR furniture applications. (Biswas, 2022) developed an AR application leveraging ARCore. Users can browse a large inventory of 3D furniture and “view objects in augmented reality, changing their size and orientation to suit their real space”. User testing showed “great accuracy and stability”, and notably “increases in satisfaction and a drop in return rates” as customers made well-informed decisions. Biswas explicitly highlighted that AR technology “offers genuine benefits to consumers and stores, greatly improving the interior design process”.

Simultaneous Localization and Mapping (SLAM) based AR systems is another such prototype. (Ramadhan, et al., 2021) applied SLAM to enable markerless indoor AR furniture placement. Their evaluation concluded that implementing SLAM “was successful” in accurately rendering furniture, and that “applying augmented reality can be considered to promote interior design products.” This suggests that robust tracking improves realism and user engagement.

To overcome early AR limitations, some studies added environmental sensing. (Alfakhori, et al., 2023) implemented reliable occlusion by using 3D environment reconstruction, tested on both indoor and outdoor models, demonstrating improved performance on mobile AR platforms. (Phade, et al., 2024) developed a lighting reconstruction framework for mobile AR which implemented guided capture and multi-rendering. These approaches demonstrate that pairing AR with pre-scanned meshes and guided capture can overcome problems of occlusion and rendering.

(Mohammed, et al., 2021) performed a detailed user experience (UX) evaluation of AR interior design applications. Using task-based testing, they identified interface issues and refined interaction methods, thereby “optimizing the user experience of AR applications”. Their work underscores the need for iterative design—asking users to perform design tasks in AR to surface usability problems.

Recommendations emerging from these studies include: use markerless AR for greater flexibility; incorporate depth sensing to improve loyalty; and employ iterative user testing to refine interfaces (Mohammed, et al., 2021). Successful prototypes (e.g., (Biswas, 2022), (Ramadhan, et al., 2021)) generally highlight the value of AR for decision support. Case studies of commercial tools (e.g., IKEA Kreativ (Nediari, et al., 2023)) mirror these findings, though they note practical issues like device support.

Theoretical Framework

A suitable theoretical framework for this research is user-centered design (UCD), which prioritizes the needs, preferences, and behaviors of end-users throughout the design and development lifecycle. Given the goal of creating an Android-based AR application for interior design clients, UCD offers a structured, iterative approach that ensures usability and accessibility are central from the onset. This is particularly important for mobile AR, where user interaction and device compatibility vary widely. Recent research supports the application of UCD in AR contexts: (Mohammed, et al., 2021) conducted a user-based evaluation of an AR interior design tool, finding that iterative feedback improved both interface clarity and user satisfaction. Similarly, (Hilken, et al., 2022) highlighted that AR tools designed with user experience in mind lead to higher engagement and decision-making confidence. UCD, additionally, directly supports the research objectives of this study, such as identifying performance limitations, designing an intuitive interface, and testing across various Android hardware profiles. By applying UCD methods (such as persona development, wireframing, and usability testing), this research ensures the application not only functions effectively, but also resonates with real user needs, ultimately enhancing client

visualization and participation in design decisions (Fleury & Chaniaud, 2024).

Question 3: Research Methodology

Research Paradigm:

The research is underpinned by the positivist paradigm, a philosophical framework that traces its epistemological foundations to the works of Auguste Comte(1929) and logical positivism of the early 20th century (Carnap, 1936). Positivism states that knowledge must be derived from empirical evidence obtained through systematic observation, measurement and experimental verification. As articulated by (Park, 2020),this paradigm prioritizes quantitative analysis and hypothesis testing, making it intrinsically suitable for this research that seeks to establish measurable relationships between variables.

In the context of this study, the positivist paradigm provides a structured framework to achieve the research objectives. The general objective —surrounding the design, development and evaluation of an AR application—demands a structured approach that is rooted in verifiable data. For instance, when addressing the specific objective of analyzing usability and performance limitations of existing AR applications on Android devices, the positivist approach enables systematic collection of numerical metrics from peer-reviewed studies, industry reports and technical evaluations. This data centric methodology as supported by (Park, 2020), ensures that conclusions about AR tool limitations are free from subjective bias thus forming a strong basis for optimization. The research questions, particularly the primary question regarding the optimization of an android-based AR tool to improve client-designer communication in interior design projects, are intrinsically suited to the positivist paradigm. Answering this question requires the measurement of quantifiable variables such as user engagement metrics, communication effectiveness scores and design comprehension rates. These numerical measurements are the cornerstone of the positivist approach, allowing us to test hypotheses and establish relationships between different elements of the AR tool and client – designer communication. By adhering to positivist principles, the study ensures that findings are rooted in empirical evidence, facilitating objective analysis and generalizable recommendations for AR tool enhancement.

Research Method:

This study adopts a quantitative, secondary research approach, which emphasizes utilizing existing numerical data drawn from peer-reviewed journals, relevant industry case studies and technical white papers. This method is suitable given that the research questions are not exploratory but evaluative, requiring direct measurement and comparison of performance and usability variables across existing AR tools. For instance, to address how AR influences client understanding compared to 2D plans, the research requires empirical metrics from controlled studies. Comparably, the assessment of hardware influence on AR experience must be built on performance benchmarks from sources such as Google ARCore technical documentation and industry standard hardware evaluation reports. High-level analysis across a range of datasets is allowed by this methodological design, while the need for resource-intensive primary experimentation—which would be constrained by the hardware fragmentation of Android and the diverse AR implementations used by existing applications—is eliminated.

The strengths of this method include efficiency in execution, strong applicability of findings and its alignment with the technological nature of the research. As (Bryman, 2012) notes, a quantitative approach is well suited for studies aiming to establish measurable relationships, which directly aligns with the objectives of the research. It allows the study to combine large volumes of data under consistent inclusion criteria allowing cross research comparison and systematic review related insights where appropriate. However, limitations exist, including variability in metric definitions across source studies, lack of access to raw experimental conditions and the potential

influence of publication bias which may distort available datasets toward favorable or innovative results. The ability of the researcher to control for unaccounted variables or test new hypotheses outside those already addressed in the literature is also suppressed by the method. Despite these limitations, the quantitative secondary approach remains the most logically and structurally appropriate for the project. It provides a disciplined data driven foundation for evaluating how android based AR tools can be refined to enhance client visualization in interior design contexts and delivers findings that are both methodologically meticulous and practically applicable within the current technological landscape.

Data Collection Methods:

This study utilizes secondary data from peer-reviewed journals, industry white papers, and technical evaluation reports published between 2020 and 2025 to analyze the effectiveness of AR in the context of interior design. Data will be collected from credible academic databases such as IEEE Xplore, ScienceDirect, and Scopus, which provides empirically validated metrics to analyze AR performance gaps and usability challenges. In addition, data will also be collected from industry white papers such as Google ARCore documentation which offers technical specifications on Android hardware limitations and optimization strategies. It will be collected from technical evaluation of AR applications such as benchmark reports which supply real-world data on client engagement and design visualization effectiveness.

The justification of the strategies and tools used for data collection quantitative data from these sources are critical to objectively address sub-questions of the study. The peer-reviewed journals provide quantitative usability metrics to address sub-question one, which focuses on usability barriers. Industry white papers and benchmark reports supply hardware-specific data to address sub-question three, which focuses on hardware variability.

To minimize bias, this research utilizes peer-reviewed journal articles published between 2020 and 2025, while only considering research that includes quantitative metrics such as usability scores, user comprehension rates, task completion times, or performance benchmarks. Meanwhile, studies that are focused exclusively on VR, PC-based AR, or unrelated industries will be excluded. To ensure the research stays objective and unbiased, multiple databases such as IEEE Xplore, ScienceDirect, Scopus, SpringerLink, Google Scholar, and Web of Science were used to avoid platform-specific bias.

To ensure the reliability of data collection, this research will implement a standardized search procedure whereby keywords such as “Augmented reality interior design”, “AR visualization effectiveness”, “AR client satisfaction”, “AR usability mobile”, “ARCore performance Android”, are used to ensure consistency.

Data Analysis Methods:

This research will apply a quantitative approach to analyze and interpret numerical data obtained from reliable published sources, following methods commonly used in recent evaluations of mobile AR usability and performance (Jalil, 2025). Descriptive statistics will be used to calculate key metrics such as means, percentages, and standard deviations, to summarize key indicators like user comprehension levels, usability scores, task duration, frame rates, and tracking errors. These statistics will be displayed clearly in structured tables and illustrative charts, providing an overview of the general patterns in how AR tools influence client visualization and interaction. (Jalil, 2025)

Comparative analysis will also be employed to assess performance results, and usability findings will be compared across various AR applications, platforms (such as ARCore and ARKit), and different Android devices. This comparative process will help identify recurring limitations and highlight differences in effectiveness among devices and systems, and results will be presented

using comparison tables and visual graphs for clarity (Harshil Bhandari, 2024).

Software tools such as Microsoft Excel will be employed to organize datasets, conduct basic statistical calculations and create visual outputs like tables and charts, while SPSS (Statistical Package for the Social Sciences) will support more advanced detailed statistical tests or cross-platform comparisons if needed. Together, this combination of tools is appropriate for handling numerical usability and performance data efficiently. Additionally, it ensures the results are presented accurately and align with the study objectives (Jalil, 2025) (Harshil Bhandari, 2024).

Handling Incomplete or Inconsistent Data:

All secondary data will be carefully checked to make sure it is completely accurate, and suitable for the analysis. To do this, data cleansing will be used; this means looking for any inconsistencies and removing them to improve the quality of the data. For example, an error could be a value that does not show the true measurement, like an incorrect frame rate or task time.

During this process, the data will be reviewed, analyzed, and any “dirty” or unreliable information will be corrected or taken out. If a source is missing important information or contains big errors, it will be left out to keep the results trustworthy. When there are small gaps in otherwise good sources, other related studies will be used to double-check or fill in missing details. All choices about changing or removing data will be written down clearly so that the whole process stays honest and transparent (Ahuja, 2024).

Question 4: Ethical Consideration

Potential Harms and Risks:

Ethical principles are fundamental guidelines that ensure the integrity and responsibility of scientific inquiry. The principles in place are not only ensure legitimate and ethically compliant research is done but to mitigate potential harm or risks towards participants of the study. The principles are considerations along the lines of obtaining informed consent as well as refraining from releasing any personal information. The principles are to be upheld throughout and beyond the research conduction and publication of the findings.

In the context of the research done, the study will solely consist of secondary data produced by previous studies that have been made public. The need for the consideration of harm and risks is reduced since no human participants will take part in the study, though there still are concerns that need to be tended to. For any data collected that has not been made public, the relevant authors will be contacted and consent to use said data will be asked for. Any data collected that contains identifiable or personal data will be stripped thereof and used strictly as data to add to the statistics. Any data collected will be used appropriately and all relevant sources will be referenced. Biases will be mitigated to avoid drawing any connections to individuals or groups by ensuring datasets are balanced and not focused on a specific demographic to prevent over- and under-representation. Data security and storage will follow best practices and protect against unauthorized access and use, the practices being storing data on secure cloud storage protected by a secure password and needing two factor authentication to gain access for authorized researchers. Research results will be reported honestly and accurately, without fabricating or misrepresenting the data. The stated ethical considerations aim to add important knowledge to research and respect the rights of the individuals and data which is included.

Ensuring research is not compromised:

All data used in this research will be obtained from credible, publicly available secondary sources such as peer-reviewed academic journals, conference proceedings, and industry white papers. To ensure data integrity and prevent any form of compromise, all sources will be stored and managed through Zotero, a secure reference management tool. The automatic backup of Zotero and version control features will be used to preserve citation accuracy and prevent data loss.

Additionally, all files will be backed up in an encrypted cloud storage, and only project team members will have access to the reference library to maintain confidentiality and prevent unauthorized alterations. As no primary or sensitive personal data is being collected, the study poses minimal risk concerning data security and privacy.

Question 5 : References

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